

INTERMEDIATE MACROECONOMICS

IS-LM MODEL OF BUSINESS CYCLES

9. LM SUBMODEL

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TWO TYPES OF ASSETS

1. **money**: used for transactions but pays no interest
 - **currency**: coins & bills
 - **checkable deposits**: funds deposited in accounts at banks and other financial institutions against which checks can be written
2. **bonds**: pay a positive rate of interest (i) but cannot be used for transaction
 - how do people allocate wealth between money & bonds?

DEMAND FOR MONEY FROM TRANSACTIONS

- people want to avoid selling bonds whenever they need money for transaction: people hold more money when they conduct more transactions
 - you have \$50,000 in wealth and spend \$3,000 a month
 - maybe you need 2 months of spending on hand
 - you keep in money: $2 \times \$3,000 = \$6,000$
 - you invest in bonds: $\$50,000 - \$6,000 = \$44,000$
 - you would keep more money if you spent more

INTEREST RATE

- bonds pay an interest rate: the higher the interest rate, the more beneficial it is to hold bonds, and the more costly it is to hold money
- bonds are held through money-market mutual funds
- in the early 1980s, the interest rate on money-market funds reached 14% per year: people moved their wealth from checking accounts to these funds to earn interest
- today interest rates are much lower (~0%) so it makes less sense to hold bonds in money-market mutual funds

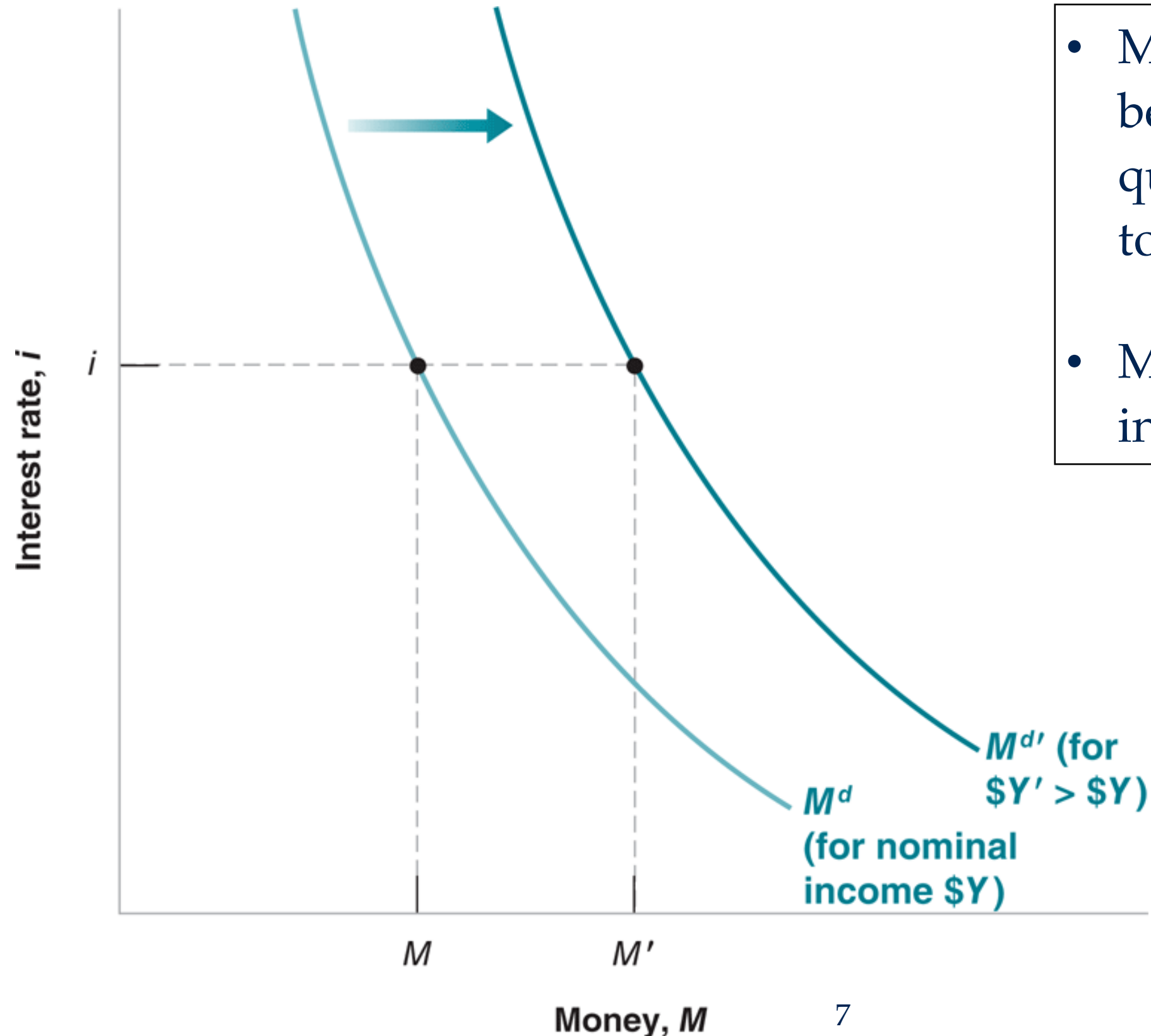
PRICE OF BONDS

- suppose a bond promises to pay \$100 a year from now
- the price of the bond today is \$P
- the interest rate i is the return on the investment of \$P
 - by definition of a rate of return: $i = (100 - P) / P$
 - equivalently, $(1+i) \times P = 100$
 - bond price and interest rate are directly related: $P = 100 / (1+i)$
- higher bond price $\longrightarrow$ lower interest rate (lower return)
- lower bond price $\longrightarrow$ higher interest rate (higher return)

THE DEMAND FOR MONEY

- money demand increases with income (Y) but decreases with interest rate (i)
 - because more income means more transactions
 - and a higher interest rate makes holding money more costly relative to bonds
- shape of money demand: $M^d(i, Y) = Y \times L(i)$ with $L'(i) < 0$
 - Y : income in the economy
 - i : interest rate on bonds
 - $L(i)$: the fraction of income that consumers hold in money to conduct transactions (decreasing in interest rate i)

THE DEMAND FOR MONEY



- M^d is a demand curve because it links a quantity demanded (M) to a price (i)
- M^d is decreasing in i but increasing in Y

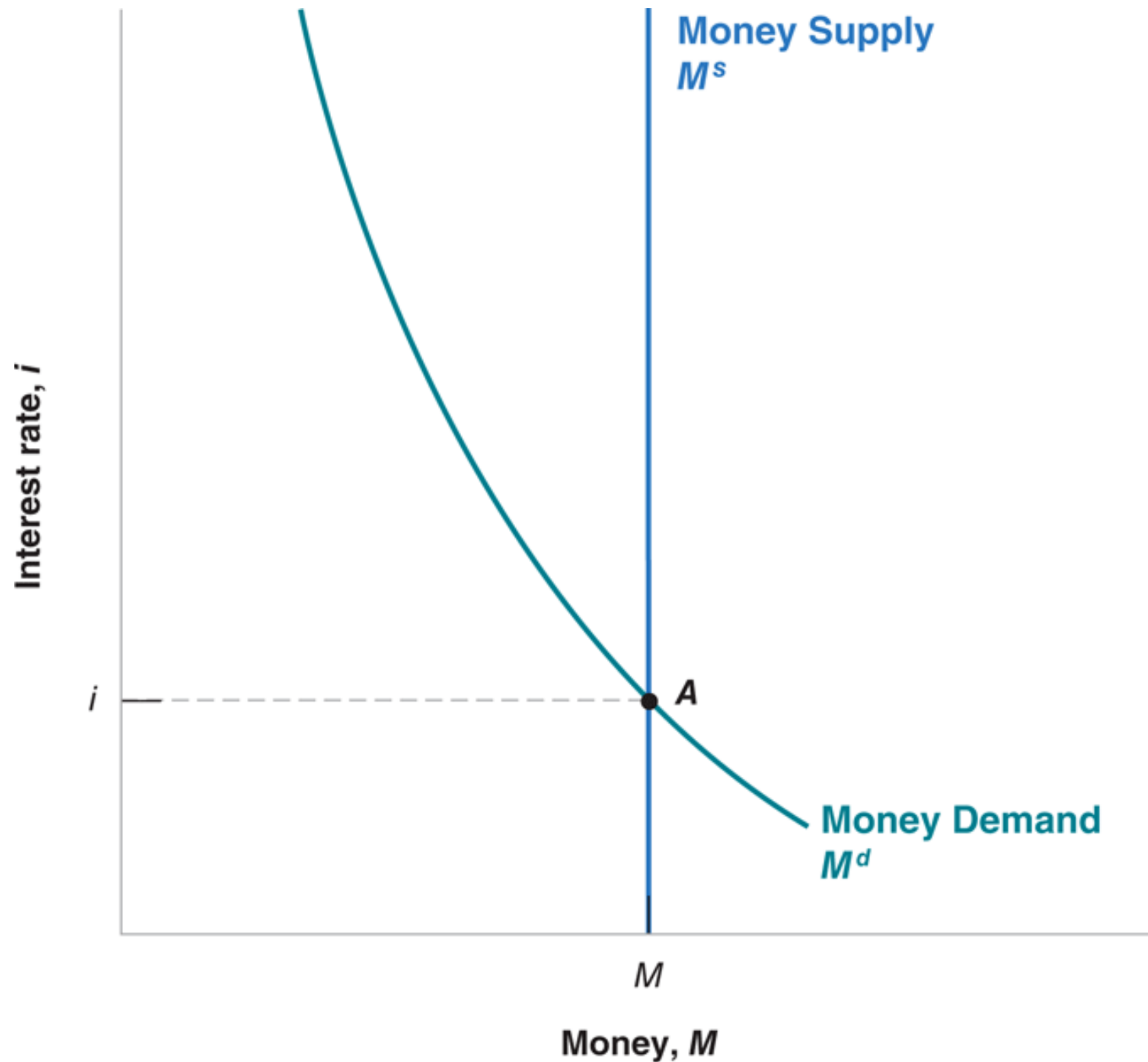
THE SUPPLY OF MONEY

- in reality, 2 types of money:
 - currency supplied by the central bank
 - checkable deposits supplied by banks
- assumption for now: the only money is currency
- the central bank supplies a quantity of money $M > 0$
 - then the money supply is $M^s = M$

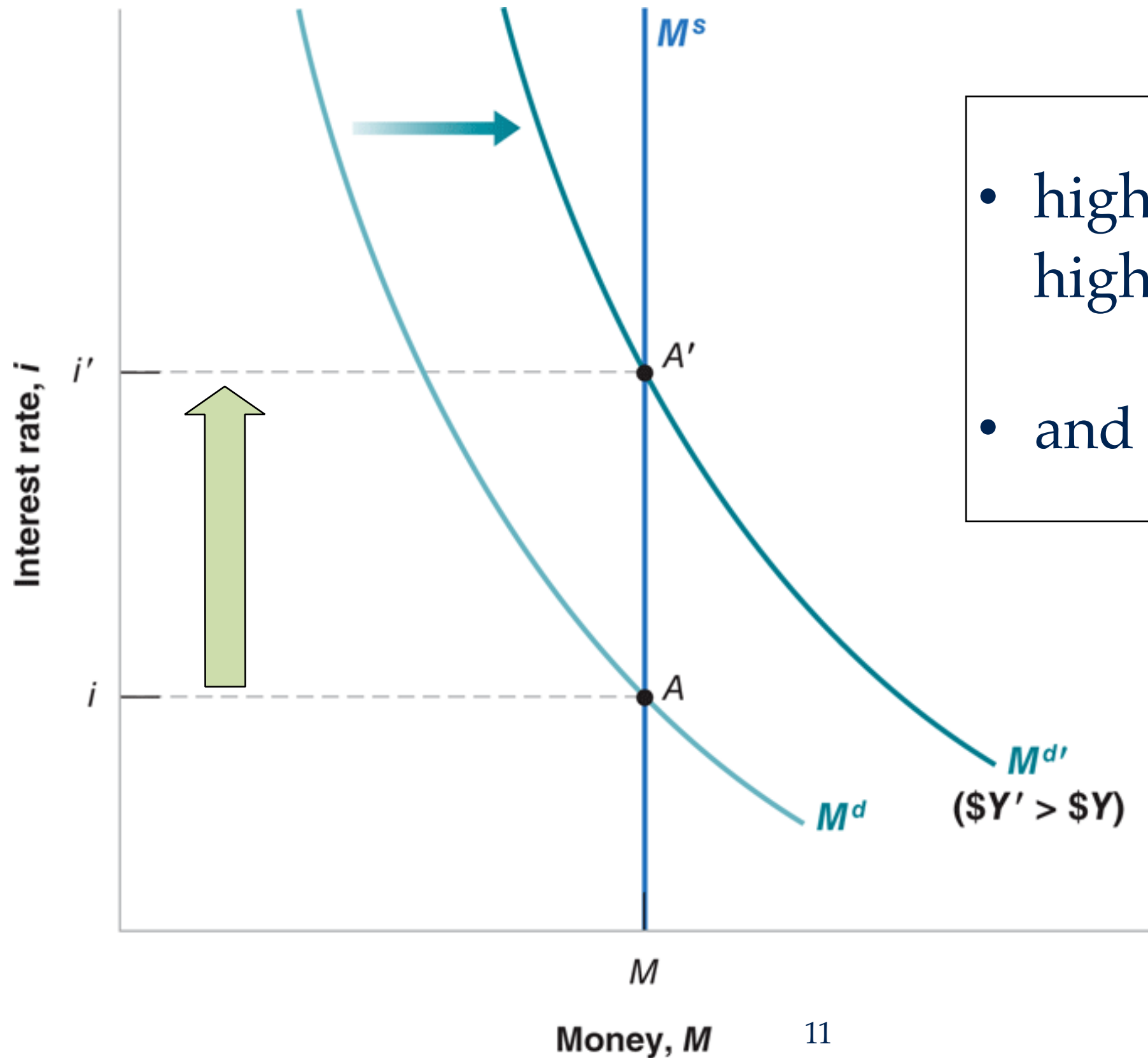
MARKET-CLEARING CONDITION

- the money market clears:
 - money supply = money demand
 - $M^s = M^d(i, Y)$ so $M = Y \times L(i)$
 - the market-clearing condition determines the interest rate i
- when the money market clears:
 - consumers are willing to absorb all the currency circulated by the central bank
 - consumers are able to get all the currency they desire

LM EQUILIBRIUM DIAGRAM

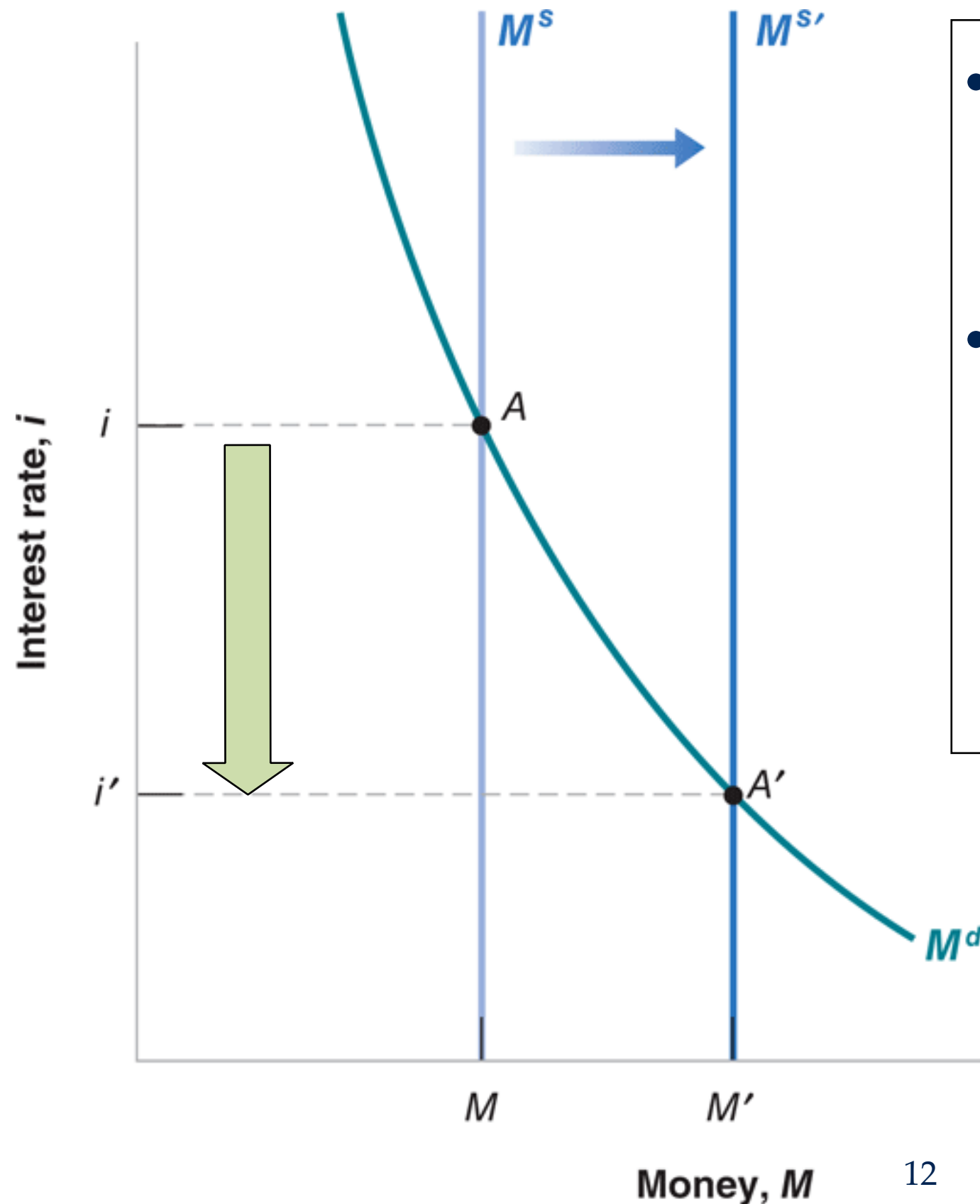


INCREASE IN INCOME



- higher income leads to higher money demand
- and higher interest rate

INCREASE IN MONEY SUPPLY



- higher money supply leads to lower interest rate
- if the interest rate did not decrease, there would be an excess supply of money: people would be unwilling to hold the extra money